

HW 5, (3)

$I \subseteq R$: proper ideal.

TFRE: (1) I is maximal

(2) R/I is a field.

Example: $\bullet (f(x)) \leq k[x]$
 $\uparrow$
irreducible

$$\bullet p\mathbb{Z} \leq \mathbb{Z}$$

$$\bullet (x, 5) \leq \mathbb{Z}[x]$$

$$\downarrow$$
$$\mathbb{Z}[x]/(x) = \mathbb{Z}$$

$$\downarrow \quad \downarrow$$
$$\mathbb{Z}[x]/(x, 5) = \mathbb{Z}/5\mathbb{Z}$$

Classifying linear transformations

Q: What are the possible linear transformations

$$T: V \longrightarrow V$$

where V is a f -dimensional k -vector space?

What do we mean by this?

$$\begin{array}{ccc} k^n & \xrightarrow{U^{-1} \circ T \circ U} & k^n \\ \cong \downarrow U & & \cong \downarrow U \Leftrightarrow (v_1, \dots, v_n) = \mathcal{B}: \\ V & \xrightarrow{T} & V \end{array} \quad \begin{array}{c} \leftarrow [T]_{\mathcal{B}} \in M_{n \times n}(k) \\ \text{Basis for } V \end{array}$$

Suppose that we also have

$$U': k^n \xrightarrow{\sim} V \Leftrightarrow (v'_1, \dots, v'_n) = \mathcal{B}' \text{ another basis}$$

$$\begin{array}{ccccc}
 k^n & \xrightarrow{\tilde{u}} & V & \xleftarrow{\tilde{u}'} & k^n \\
 \downarrow [T]_{\mathcal{B}} & & \downarrow \tau & & \downarrow [T]_{\mathcal{B}'} \\
 k^n & \xrightarrow[u]{\tilde{u}} & V & \xleftarrow[u']{\tilde{u}'} & k^n
 \end{array}$$

$$\begin{array}{ccc}
 & \xrightarrow{C^{-1}} & \\
 k^n & \xrightarrow[(u')^{-1} \circ u]{\tilde{u}} & k^n \\
 \downarrow [T]_{\mathcal{B}} & \xrightarrow{C \in GL_n(k)} & \downarrow [T]_{\mathcal{B}'} \\
 k^n & \xrightarrow[(u')^{-1} \circ u]{\tilde{u}} & k^n \\
 & \xrightarrow{C} &
 \end{array}$$

We find:

$$[T]_{\mathcal{B}'} = C [T]_{\mathcal{B}} C^{-1}$$

for $C \in GL_n(k)$

Defn $A_1, A_2 \in M_{n \times n}(k)$ are conjugate
if $\exists C \in GL_n(k)$ s.t. $CA_1C^{-1} = A_2$

Rmd: $GL_n(k) \hookrightarrow M_{n \times n}(k)$

$$(C, A) \mapsto CAC^{-1}$$

This is the Conjugation action

A_1 is conjugate to A_2

$(\Leftrightarrow)$ A_1, A_2 are in the same orbit

Fact: $T: V \rightarrow V$

$$u, u': k^n \xrightarrow{\sim} V$$

$\downarrow \quad \downarrow$
 $B \quad B'$

bases

Then $[T]_B$ & $[T]_{B'}$

are conjugate